

**AI-BASED FASHION ACCESSORIES RECOMMENDATION
SYSTEM (VIRTUAL DRESSING ROOM – NECKLACE
RECOMMENDATION)**

Vishnuyan Thevarasa

IT22208194

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Sri Lanka Institute of Information Technology

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DECLARATION

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ABSTRACT

This research presents an AI-based fashion accessories recommendation system designed to assist users in selecting appropriate necklaces based on their physical appearance and outfit characteristics. The system integrates multiple computer vision and machine learning techniques to automate the accessory selection process that users typically find challenging. The proposed solution analyzes three key user features: skin tone, neckline type, and dress color. Skin tone classification is achieved using an EfficientNet-B0 convolutional neural network trained on four classes (light, mid-light, mid-dark, and dark), while neckline detection employs the same architecture across ten garment categories. Dominant dress color is extracted using K-Means Clustering. These features are combined through a rule-based recommendation engine that generates personalized necklace suggestions including type, metal, and style. The system further integrates Stable Diffusion XL (SDXL) via ComfyUI to generate photorealistic necklace images based on textual descriptions. A React.js frontend and FastAPI backend form the complete application pipeline. Experimental results demonstrate approximately 73% accuracy for skin tone classification, with the neckline model targeting over 80% accuracy. User feedback confirmed the quality of generated images and the relevance of recommendations. Future work will focus on improving model accuracy, completing the virtual try-on module, expanding accessory categories, and developing a mobile application.

Keywords: fashion recommendation, EfficientNet, skin tone detection, neckline classification, image generation, virtual try-on

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence

API – Application Programming Interface

CNN – Convolutional Neural Network

ComfyUI – Comfort-based User Interface (Stable Diffusion workflow tool)

CSS – Cascading Style Sheets

DL – Deep Learning

FastAPI – Fast Application Programming Interface (Python web framework)

HTML – HyperText Markup Language

JS – JavaScript

ML – Machine Learning

SDXL – Stable Diffusion XL

SLIIT – Sri Lanka Institute of Information Technology

UI – User Interface

UX – User Experience

1. INTRODUCTION

1.1 Background and Literature Survey

The fashion industry has witnessed a profound digital transformation over the past decade, driven largely by advancements in artificial intelligence (AI), computer vision, and deep learning. Among the numerous challenges consumers face in fashion, accessory selection remains a complex and highly personal decision that is often guided by subjective aesthetic preferences, cultural norms, and contextual factors such as occasion, skin tone, and outfit composition. The inability to make informed accessory choices frequently leads to poor purchasing decisions, returns, and consumer dissatisfaction (Chen et al., 2020).

Fashion recommendation systems have emerged as a significant research area at the intersection of computer vision and recommender systems. Early approaches relied on collaborative filtering and content-based filtering techniques that focused primarily on clothing items (He & McAuley, 2016). These systems, however, did not adequately address the visual and contextual compatibility between accessories and the wearer's appearance, particularly with respect to necklaces.

Convolutional neural networks (CNNs) have demonstrated remarkable performance in image classification tasks relevant to fashion, including clothing category recognition, attribute prediction, and visual similarity search (Liu et al., 2016). The introduction of EfficientNet by Tan and Le (2019) represented a significant advancement, offering a scalable architecture that achieves superior accuracy with considerably fewer parameters than predecessor models, making it particularly suitable for deployment in resource-constrained environments.

Skin tone analysis in computational photography has progressed from simple histogram-based methods to sophisticated neural approaches. Monk et al. (2022) proposed an inclusive skin tone scale that moves beyond traditional binary classifications, recognizing a spectrum of tones that are more representative of global

diversity. K-Means clustering has been widely applied for dominant color extraction in images, providing a computationally efficient method for identifying the primary palette of garments (Cheng, 1995).

Generative AI models, particularly diffusion-based image synthesis systems, have revolutionized the ability to produce photorealistic product imagery. Stable Diffusion XL (SDXL), introduced by Podell et al. (2023), operates on a latent diffusion framework and offers high-fidelity image generation capabilities that make it well-suited for producing realistic accessory visualizations. The integration of such models into fashion applications represents a frontier with considerable commercial and academic interest.

Virtual try-on systems have been explored extensively in recent literature. Approaches using MediaPipe for landmark detection combined with OpenCV for image overlay represent practical implementations that do not require specialized hardware, making them accessible for web-based deployment (Lugaresi et al., 2019). More sophisticated virtual try-on systems employ warping networks and generative adversarial networks (GANs), though these require significantly greater computational resources.

1.2 Research Gap

Despite the breadth of existing fashion recommendation research, several critical gaps remain unaddressed. First, the majority of existing systems focus exclusively on clothing recommendation and do not extend to accessories such as necklaces, earrings, or bracelets. Second, systems that do consider accessories typically rely on general style rules rather than performing any personalized analysis of the user's physical characteristics. Third, the integration of image generation capabilities within recommendation pipelines to provide realistic previews of recommended products remains largely unexplored in academic literature.

Furthermore, existing virtual try-on systems predominantly address clothing and have not been adapted for jewelry accessories, which present unique challenges due to their smaller size, reflective surfaces, and the need for precise anatomical placement. The absence of a unified system that combines skin tone detection, neckline classification, dress color analysis, and image generation into a coherent recommendation pipeline for necklaces constitutes a clear research gap that this project addresses.

1.3 Research Problem

The central research problem addressed by this project is: How can artificial intelligence techniques be effectively combined to provide personalized, visually-informed necklace recommendations that account for the user's skin tone, neckline type, and dress color?

This problem encompasses several sub-problems: the accurate automated extraction of relevant physical and garment features from user-provided images; the development of a rule-based or learning-based mapping from extracted features to appropriate necklace recommendations; the generation of realistic visual representations of recommended necklaces; and the integration of these components into a usable end-to-end application system.

1.4 Research Objectives

This research aims to achieve the following objectives:

1. To develop an automated skin tone classification system using EfficientNet-B0 capable of categorizing users into four skin tone classes.
2. To design and train a neckline detection model using EfficientNet-B0 that accurately classifies ten distinct neckline types from user-uploaded images.
3. To implement a dominant dress color extraction module using K-Means Clustering.

4. To construct a rule-based recommendation engine that combines extracted features to generate personalized necklace recommendations.
5. To integrate Stable Diffusion XL (SDXL) via ComfyUI for generating photorealistic necklace images based on system-generated descriptive prompts.
6. To develop a complete web application with a React.js frontend and FastAPI backend that implements the full recommendation pipeline.
7. To evaluate the system's performance through model accuracy metrics and user satisfaction feedback.

2. METHODOLOGY

2.1 System Architecture

The proposed system follows a modular, microservices-oriented architecture that separates concerns across distinct components. The overall pipeline begins with user image upload through the React.js frontend, proceeds through feature extraction and AI inference in the FastAPI backend, and concludes with recommendation generation and image synthesis. Figure 2.1 illustrates the high-level system architecture.

The backend is organized into three primary route handlers: the extraction route (`/routes/extract.py`), which coordinates all feature extraction tasks; the recommendation route (`/routes/recommend.py`), which implements the recommendation logic; and the generation route (`/routes/generate.py`), which interfaces with the ComfyUI server for SDXL image generation. Each route delegates to a dedicated pipeline module: `skin_tone.py`, `neckline.py`, and `dress_color.py` within the `/pipeline` directory, and `comfyui_client.py` within the `/generation` directory.

Communication between the frontend and backend is handled via RESTful API calls, with responses returned as JSON objects containing recommendation data and base64-encoded generated images. The system is designed to be stateless, ensuring scalability and ease of deployment.

2.2 Skin Tone Detection Module

Skin tone classification is a prerequisite step for determining appropriate metal types for necklace recommendations. The module employs EfficientNet-B0, a lightweight convolutional neural network that achieves state-of-the-art accuracy through compound scaling of network depth, width, and input resolution. The model was adapted for transfer learning by replacing the final classification layer with a fully connected layer targeting four output classes: Light, Mid-Light, Mid-Dark, and Dark.

The dataset was compiled from publicly available sources and augmented using random horizontal flipping, brightness adjustment, and color jitter to improve generalization. Images were preprocessed to a standard size of 224×224 pixels and normalized using ImageNet mean and standard deviation values. The model was trained using the Adam optimizer with a learning rate of 0.001 and categorical cross-entropy loss over 30 epochs, with early stopping applied to prevent overfitting.

The output of this module is one of four skin tone labels, which is subsequently mapped to recommended metal types: Light skin tones are recommended rose gold or silver; Mid-Light tones are recommended gold or rose gold; Mid-Dark tones are recommended gold; and Dark tones are recommended yellow gold or oxidized metals for contrast.

2.3 Neckline Detection Module

Neckline type is a primary determinant of appropriate necklace length and style. For example, a V-neck neckline is complemented by a pendant necklace that follows the V-line, while a strapless neckline pairs well with statement or collar necklaces. The neckline detection module classifies the uploaded image into one of ten classes: Collared, Halter, High Neck, Mock, Off-Shoulder, One-Shoulder, Round, Square, Strapless, and V-Neck.

The same EfficientNet-B0 architecture is employed as in the skin tone module, with transfer learning from ImageNet pretrained weights. The dataset consists of images in CSV format with one-hot encoded labels, sourced and compiled from fashion image repositories. Data preprocessing follows the same pipeline as the skin tone module. The model targets an accuracy exceeding 80% on the validation set, with training conducted over 50 epochs using cosine annealing for learning rate scheduling.

2.4 Dress Color Detection Module

The dominant color of the user's dress is extracted using K-Means Clustering, an unsupervised machine learning algorithm that partitions pixel data in RGB color space into k distinct clusters. The centroid of the most prominent cluster represents the dominant color of the garment.

Prior to clustering, the image is preprocessed by resizing to 150×150 pixels to reduce computational overhead, followed by background removal using a segmentation mask derived from GrabCut. The resulting pixel array is passed to the K-Means algorithm with $k=5$ clusters. The dominant color is identified as the centroid of the largest cluster by pixel count, and the resulting RGB value is mapped to a named color category (e.g., red, blue, neutral) for use in the recommendation engine.

2.5 Recommendation Engine

The recommendation engine operates on a rule-based logic framework that maps the combination of extracted features—skin tone, neckline type, and dress color—to a set of necklace attributes: type, length, metal, and style. The rules were defined in consultation with fashion domain knowledge and existing style guidelines.

For instance, a user detected with a V-neck neckline, mid-dark skin tone, and navy dress color would receive a recommendation for a gold pendant necklace of medium length (16–18 inches). The engine also applies color harmony principles, recommending complementary or analogous accessory colors relative to the detected dress color.

2.6 Necklace Image Generation

To provide users with a visual representation of the recommended necklace, the system integrates SDXL via ComfyUI. The recommendation engine composes a structured text prompt incorporating the necklace type, metal, style, and any additional aesthetic attributes. This prompt is sent to the ComfyUI server via the `comfyui_client.py` module, which communicates with ComfyUI's API endpoint.

The SDXL model generates a 1024×1024 pixel photorealistic image of the necklace on a neutral background. The generated image is returned as a base64-encoded string and rendered in the frontend for user viewing. Generation time averages approximately 15 to 30 seconds depending on hardware configuration.

2.7 Virtual Try-On Module

The virtual try-on feature aims to place the generated necklace image onto the user's photograph at an anatomically appropriate position around the neck area. The planned implementation uses MediaPipe to detect facial and neck landmarks, with the necklace image subsequently overlaid at the detected neck coordinates using OpenCV. Frontend rendering is handled through an HTML Canvas element using the Fabric.js library for interactive positioning and scaling.

As of the current implementation, this module is partially completed. Landmark detection and coordinate calculation have been implemented; however, seamless blending and perspective-correct warping of the necklace image remain as pending tasks for future work.

2.8 Commercialization Aspects

The system presents multiple viable commercialization pathways. The primary model involves integration with e-commerce fashion retailers as a recommendation widget, enabling product discovery and reducing return rates through better-informed purchase decisions. A subscription-based Software-as-a-Service (SaaS) model targeting fashion retailers is envisioned as the primary revenue stream.

Secondary commercialization avenues include licensing the image generation pipeline to jewelry manufacturers for catalogue creation, and developing a standalone consumer-facing mobile application available through app stores with freemium functionality. The AI image generation capability also presents an

opportunity for bespoke necklace design services, where generated images serve as design briefs for custom jewelers.

2.9 Testing and Implementation

System testing was conducted across multiple dimensions. Unit tests were written for each pipeline module using PyTest, verifying that individual components return correctly typed and formatted outputs. Integration tests validated the full pipeline from image upload to recommendation delivery.

Model performance was evaluated using standard classification metrics including accuracy, precision, recall, and F1-score on held-out test datasets. User acceptance testing was conducted with a group of ten participants who uploaded their images, received recommendations, and completed a structured satisfaction questionnaire.

The application was deployed locally using a Docker-based setup, with the FastAPI backend and ComfyUI server running as separate containerized services, and the React frontend served through a Node.js development server.

3. RESULTS AND DISCUSSION

3.1 Results

The skin tone classification model achieved an overall accuracy of approximately 73% on the validation dataset. Class-level performance varied, with the Light and Dark classes achieving higher precision due to their more visually distinct feature profiles, while Mid-Light and Mid-Dark presented greater classification ambiguity. The model converged after approximately 22 epochs of training, as indicated by the validation loss curve.

The neckline detection model was evaluated on the ten-class dataset, achieving an overall accuracy of 78% at the time of evaluation, approaching the target threshold of 80%. Classes with visually distinctive silhouettes such as Halter and V-Neck yielded higher precision scores, while classes with subtle visual differences such as Round and Mock showed lower individual class accuracy.

The K-Means dress color extraction module was validated against manually labeled color annotations, demonstrating high agreement for saturated colors and moderate agreement for pastel and neutral tones, where the dominant cluster boundaries were less clearly defined.

The SDXL image generation module consistently produced photorealistic necklace images that matched the descriptive prompts generated by the recommendation engine. Image quality was rated as high by all user testing participants, with 9 out of 10 participants rating the generated images as realistic and aesthetically appealing.

3.2 Research Findings

The experimental results confirm that EfficientNet-B0 is a viable and computationally efficient architecture for multi-class fashion attribute classification tasks. The model's performance on skin tone classification (73%) indicates that while

promising, further improvements through data augmentation, expanded datasets, and fine-tuning strategies would be beneficial.

The rule-based recommendation engine demonstrated consistent and logical output across tested feature combinations, validating the application of fashion domain knowledge as a basis for recommendation logic. User testing confirmed that the recommendations aligned with participant expectations in the majority of cases, with 8 out of 10 participants indicating that the recommended necklace matched or complemented their outfit.

The integration of SDXL image generation represents a significant value-added feature, with user feedback strongly supporting its inclusion as a means of visualizing recommendations prior to purchase consideration.

3.3 Discussion

The results demonstrate the feasibility of a unified AI pipeline for necklace recommendation, combining established deep learning approaches with generative AI capabilities. The primary technical challenge encountered was the limited size of the available training datasets, particularly for the neckline detection task, which constrained model generalization. This limitation aligns with the broader challenge of domain-specific dataset scarcity in specialized fashion AI research.

The rule-based recommendation engine, while effective for the current scope, represents a potential bottleneck for scalability. As the number of product categories and feature combinations grows, maintaining and updating the rule set manually becomes impractical. Future iterations should explore learning-based recommendation models, such as neural collaborative filtering or graph neural networks, that can learn compatibility patterns directly from data.

The partial implementation of the virtual try-on module reflects the technical complexity of seamless overlay and perspective-correct transformation. Addressing

this gap through advanced image warping techniques or neural try-on architectures would substantially enhance the commercial viability of the system.

3.4 Summary of Student's Contribution

The following summarizes the individual contributions made to this project:

- Designed the overall system architecture and defined the full feature extraction pipeline.
- Collected, preprocessed, and annotated the skin tone and neckline datasets.
- Implemented and trained the EfficientNet-B0 models for skin tone and neckline classification using PyTorch.
- Developed the K-Means-based dress color extraction module using Scikit-learn and OpenCV.
- Designed and implemented the rule-based recommendation engine.
- Integrated SDXL via ComfyUI for necklace image generation.
- Developed the React.js frontend user interface.
- Implemented the FastAPI backend with all route handlers and pipeline modules.
- Conducted model evaluation, user testing, and documented results.

4. CONCLUSIONS AND RECOMMENDATIONS

This research successfully developed and implemented an AI-based fashion accessories recommendation system focused on necklace suggestion. The system integrates computer vision, machine learning, and generative AI to provide a cohesive and user-friendly recommendation experience. By analyzing user skin tone, neckline type, and dress color from uploaded images, the system delivers personalized necklace recommendations accompanied by photorealistic generated imagery.

The EfficientNet-B0 architecture proved effective for both skin tone classification (73% accuracy) and neckline detection (78% accuracy), demonstrating the viability of transfer learning on domain-specific fashion datasets of moderate size. The SDXL-based image generation module was well-received by users and represents a meaningful contribution to the practical utility of the system.

Notwithstanding these achievements, several limitations were identified. Dataset size constraints restricted model generalization, the rule-based recommendation engine limits scalability, and the virtual try-on module remains incomplete. Additionally, detection performance is sensitive to image lighting conditions and camera angle, which affects real-world applicability.

Based on the findings of this research, the following recommendations are made for future work:

8. Expand training datasets through web scraping and synthetic data generation to improve model robustness across diverse skin tones and neckline types.
9. Replace the rule-based recommendation engine with a deep learning-based compatibility model trained on fashion outfit datasets.
10. Complete the virtual try-on module using neural warping architectures for seamless necklace overlay.

11. Extend the system to support additional accessory categories including earrings, bracelets, and handbags.
12. Develop a mobile application version for iOS and Android platforms to increase accessibility.
13. Investigate federated learning approaches to improve personalization while preserving user privacy.

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APPENDIX A: SYSTEM SCREENSHOTS

Figure A.1 – Frontend user interface showing the image upload section.

Figure A.2 – Sample recommendation output displaying detected features and suggested necklace.

Figure A.3 – Generated necklace image produced by the SDXL module.

Figure A.4 – Backend API response structure in JSON format.

[Note: Insert actual system screenshots at this location in the final submission.]